SRI CHANDRASEKHARENDRA SARASWATHI VISWA MAHAVIDYALAYA (SCSVMV)

(Deemed to be University u/s 3 of the UGC Act 1956)

(Accredited with 'A' Grade by NAAC)

ENATHUR, KANCHIPURAM –631 561

WEB TECHNOLOGY LAB

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BONAFIDE CERTIFICATE

This is to certify that this is the Bonafide record of work done by Ms. B MADHUSHREE with Reg.No. 112413013 of III-Year B.Sc., (COMPUTER SCIENCE WITH DATA SCIENCE) in the WEB TECHNOLOGY Lab during the year 2026.

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Submitted for the Practical Examination held on 16-09-2026

Internal Examiner **External Examiner**

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1	**SIMPLE HTML: Login Form**	29Jun2026

AIM:

To familiarize students with the basic HTML tags: Design a simple Login Form

CONTEXT:

HTML (Hypertext Markup Language) is the standard markup language used to create and structure web pages. It uses tags and elements to define the content and structure of a webpage.

Tag	Description
<html>	Defines the HTML document.
<head>	Contains information about the webpage.
<title>	Defines the title displayed on the browser tab.
<body>	Contains the visible content of the webpage.
<h2>	Defines a heading.
<form>	Defines a form for collecting user input.
<label>	Defines a label for an input field.
<input>	Creates input fields such as email and password fields.
<button>	Creates a clickable button for submitting the form.

SOURCE CODE:

login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
</head>
<body>
<center>
<h1>Login</h1>
<form>
   <label>Email ID:</label>
   <input type="email" name="email">
   <br><br>
   <label>Password:</label>
   <input type="password" name="password">
   <br><br>
   <input type="submit" value="Login">
   <input type="reset" value="Clear">
</form>
</center>
</body>
</html>
```

OUTPUT:



RESULT:

Thus, a simple Login Form was successfully designed using HTML and displayed in the web browser.

2	**INPUT FIELDS: Student Registration Form**	06Jul2026

AIM:

To familiarize students with HTML form elements and input fields and create a Student Registration Form for collecting student details.

CONTEXT:

HTML forms are used to collect information from users through different form controls. Input fields allow users to enter or select data such as personal details, contact information, gender, course, year, address, and password. Tags and input types used in the program:

Tag Input Type	**Description**
<form>	Defines a form for collecting user information.
<label>	Defines a label for an input field.
<input>	Creates an input field.
type="text"	Creates a field for entering text.
type="date"	Creates a field for selecting a date.
type="radio"	Creates radio buttons for selecting an option.
type="email"	Creates a field for entering an email address.
type="tel"	Creates a field for entering a telephone number.
type="password"	Creates a password input field.
type="submit"	Creates a button to submit the form.
type="reset"	Creates a button to clear/reset the form.
<select>	Creates a drop-down list.
<option>	Defines an option in a drop-down list.
<textarea>	Creates a multi-line text input area.
 	Inserts a line break.

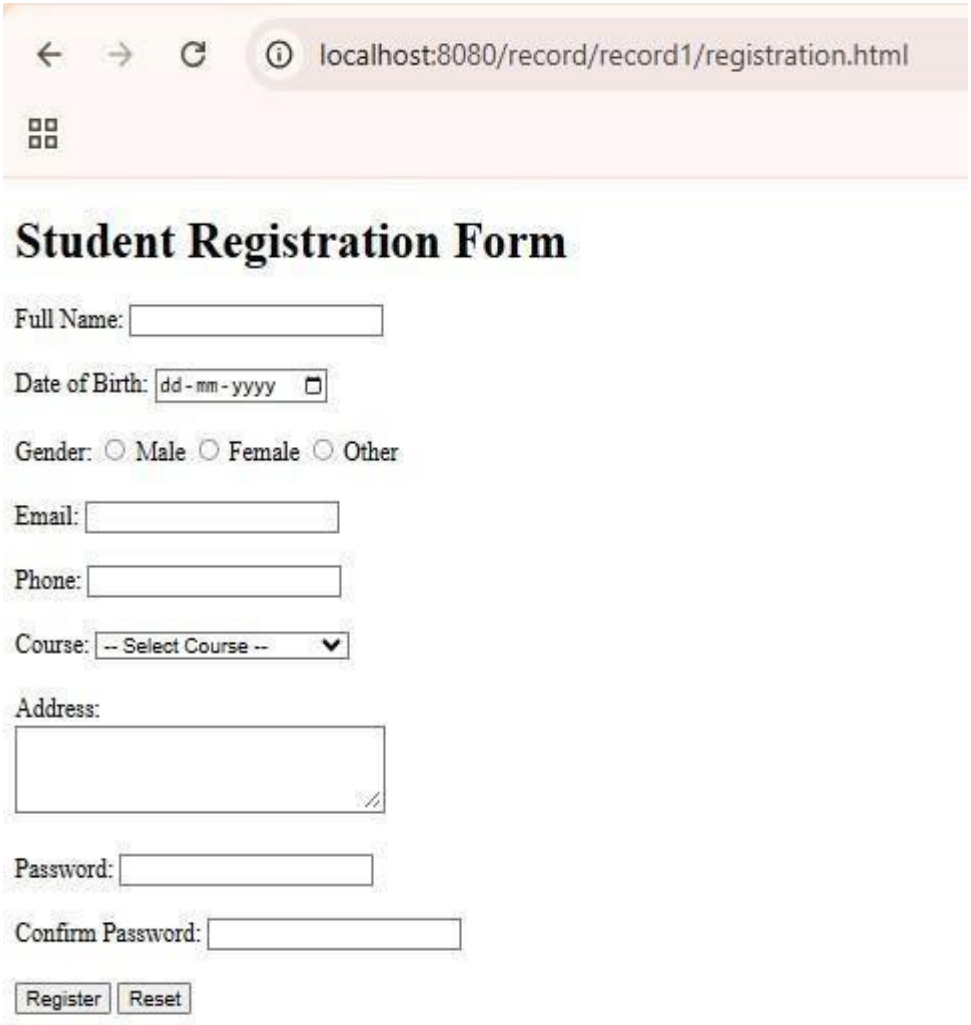
SOURCE CODE:

registration.html

```
<!DOCTYPE html>
    <html lang="en">
    <head>
      <meta charset="UTF-8">
      <title>Student Registration</title>
    </head>
    <body>
      <h1>Student Registration Form</h1>
      <form>
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name">
        <br><br>
        <label for="dob">Date of Birth:</label>
        <input type="date" id="dob" name="dob">
        <br><br>
        <label>Gender:</label>
        <input type="radio" name="gender" value="Male"> Male
        <input type="radio" name="gender" value="Female"> Female
        <input type="radio" name="gender" value="Other"> Other
        <br><br>
        <label for="email">Email:</label>
        <input type="email" id="email" name="email">
        <br><br>
        <label for="phone">Phone:</label>
        <input type="tel" id="phone" name="phone">
        <br><br>
        <label for="course">Course:</label>
        <select id="course" name="course">
          <option value="">-- Select Course --</option>
          <option>BCA</option>
          <option>B.Sc Computer Science</option>
          <option>B.Sc IT</option>
          <option>B.Com</option>
        </select>
        <br><br>
        <label for="address">Address:</label>
        <br>
        <textarea id="address" name="address" rows="3" cols="30"></textarea>
        <br><br>
        <label for="password">Password:</label>
        <input type="password" id="password" name="password">
        <br><br>
```

```
<label for="confirmPassword">Confirm Password:</label>
<input type="password" id="confirmPassword" name="confirmPassword">
<br><br>
<input type="submit" value="Register">
<input type="reset" value="Reset">
</form>  </body></html>
```

OUTPUT:



RESULT:

Thus, a Student Registration Form was successfully designed using HTML with various input fields, form controls, and buttons.

3	**APPLYING CSS: Student Login Using Html and CSS**	13Jul2026

AIM:
To familiarize students with CSS properties used for styling web pages and to design a simple Login Form using HTML and CSS.

CONTEXT:
CSS (Cascading Style Sheets) is used to control the appearance and layout of elements on a webpage. It allows properties such as colors, fonts, spacing, borders, alignment, and dimensions to be applied to webpage elements.

In this program, CSS is used to style the Login Form by centering the login box, setting its background and border, formatting the input fields, and styling the Submit and Reset buttons. Flexbox properties are used to position the login box at the center of the page.

CSS Property	Purpose
display: flex	Enables flexible layout for positioning elements.
justify-content	Aligns content horizontally.
align-items	Aligns content vertically.
background-color	Sets the background color.
font-family	Specifies the font style.
width	Sets the width of an element.
padding	Adds space inside an element.
margin	Adds space outside an element.
border	Adds a border around an element.
border-radius	Creates rounded corners.
text-align	Aligns text within an element.
color	Sets the text color.

SOURCE CODE:

login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login</title>
   <link rel="stylesheet" href="login.css">
</head>
<body>
   <div class="login-box">
      <h1>Login</h1>

      <form>
         <div class="form-row">
            <label>Email ID:</label>
            <input type="email" name="email" required>
         </div>
         <div class="form-row">
            <label>Password:</label>
            <input type="password" name="password" required>
         </div>
         <div class="buttons">
            <button type="submit">Login</button>
            <button type="reset">Clear</button>
         </div>
      </form>
   </div>
</body>
</html>
```

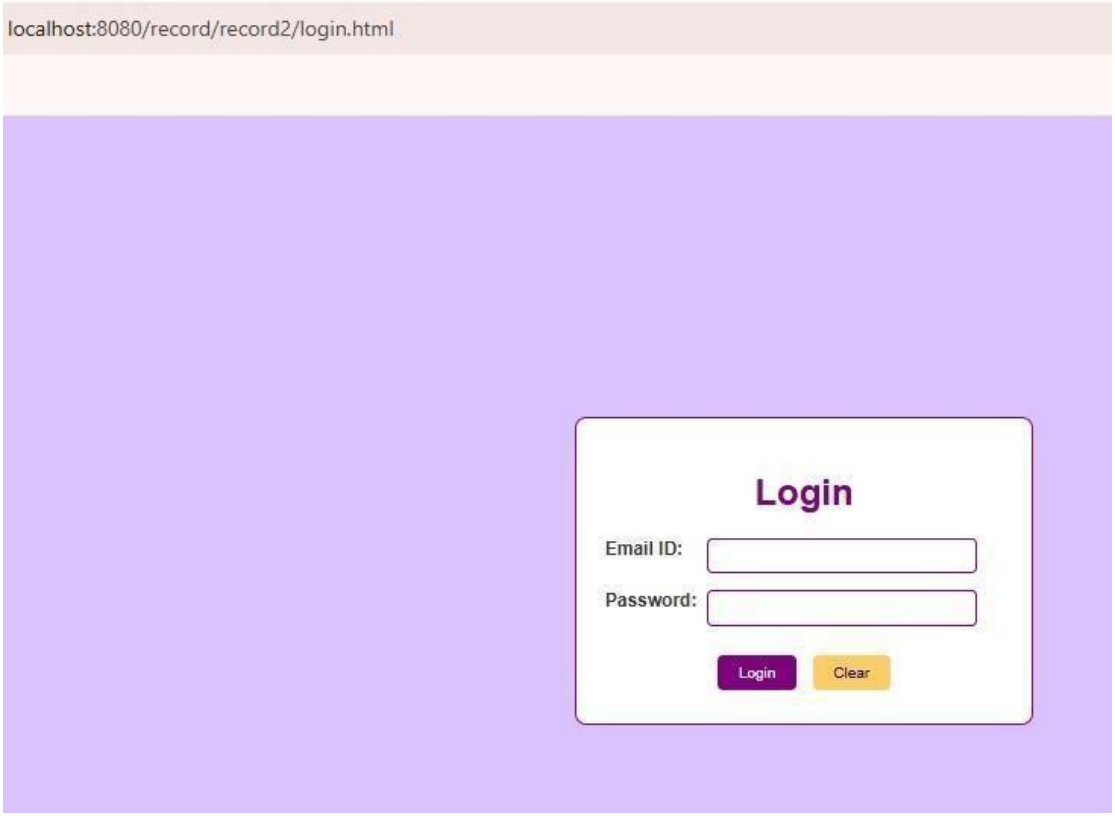
login.css

```
body {
   margin: 0;
   height: 100vh;
   display: flex;
   justify-content: center;
   align-items: center;
   font-family: Arial;
   background-color: #d9c2ff;
}
.login-box {
   width: 350px;
   padding: 25px;
   background-color: white;
   border: 2px solid purple;
```

```
    border-radius: 10px;
}
h1 {
    text-align: center;
    color: purple;
}
.form-row {
    display: flex;
    margin: 15px 0;
}
label {
    width: 90px;
    font-weight: bold;
  color: #333;
}
input {
    width: 220px;
    padding: 7px;
    border: 1px solid purple;
    border-radius: 5px;
}
.buttons {
    text-align: center;
    margin-top: 20px;
}
button {
    padding: 8px 18px;
    margin: 5px;
    border: none;
    border-radius: 5px;
}
button[type="submit"] {
    background-color: purple;
    color: white;
}
button[type="reset"] {
    background-color: #ffcc66;
}
```

OUTPUT:

Positive Scenario – Login Form with CSS Styling



Negative Scenario – Login Form without CSS Styling



RESULT:

Thus, a simple Login Form was successfully designed using HTML and styled using CSS with proper alignment, spacing, colors, borders, and button formatting.

4	**APPLYING CSS: Registration Form Using Html/CSS**	22Jul2026

AIM:

To familiarize students with different HTML form input types and form elements, and to design a Student Registration Form using HTML and CSS.

CONTEXT:

The Student Registration Form uses different form elements and input types to collect student details. CSS properties are used to improve the appearance and layout of the form.

HTML Tags and Input Types Used:

Tag / Input Type	Purpose
<form>	Defines the registration form.
<label>	Defines labels for input fields.
<input>	Creates different input fields.
type="date"	Allows selection of date of birth.
type="radio"	Allows selection of one option.
<select>	Creates a drop-down list.
<textarea>	Creates a multi-line input field.
type="password"	Creates a password field.

CSS Properties Used:

CSS Property	Purpose
background-color	Sets the background color.
width	Sets the width of an element.
padding	Adds space inside an element.
margin	Adds space outside an element.
border	Defines the border.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow effect.
text-align	Aligns text.

SOURCE CODE:

registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Registration</title>
   <link rel="stylesheet" href="registration.css">
</head>
<body>
   <h1>Student Registration Form</h1>
   <p>Please enter your details to create your student account.</p>
<form>
      <label for="name">Full Name:</label>
      <input type="text" id="name" name="name">
      <br><br>
      <label for="dob">Date of Birth:</label>
      <input type="date" id="dob" name="dob">
      <br><br>
      <label>Gender:</label>
      <input type="radio" name="gender" value="Male"> Male
      <input type="radio" name="gender" value="Female"> Female
      <input type="radio" name="gender" value="Other"> Other
      <br><br>
      <label for="email">Email:</label>
      <input type="email" id="email" name="email">
      <br><br>
      <label for="phone">Phone:</label>
      <input type="tel" id="phone" name="phone">
      <br><br>
      <label for="course">Course:</label>
      <select id="course" name="course">
         <option value="">-- Select Course --</option>
         <option>BCA</option>
         <option>B.Sc Computer Science</option>
         <option>B.Sc IT</option>
         <option>B.Com</option>
      </select>
       <br><br>
      <label for="address">Address:</label>
      <br>
      <textarea id="address" name="address" rows="3" cols="30"></textarea>
<br><br>
```

```
            <label for="password">Password:</label>
            <input type="password" id="password" name="password">
            <br><br>
            <label for="confirmPassword">Confirm Password:</label>
            <input type="password" id="confirmPassword" name="confirmPassword">
            <br><br>
            <input type="submit" value="Register">
            <input type="reset" value="Reset">
        </form>
    </body>
    </html>
```

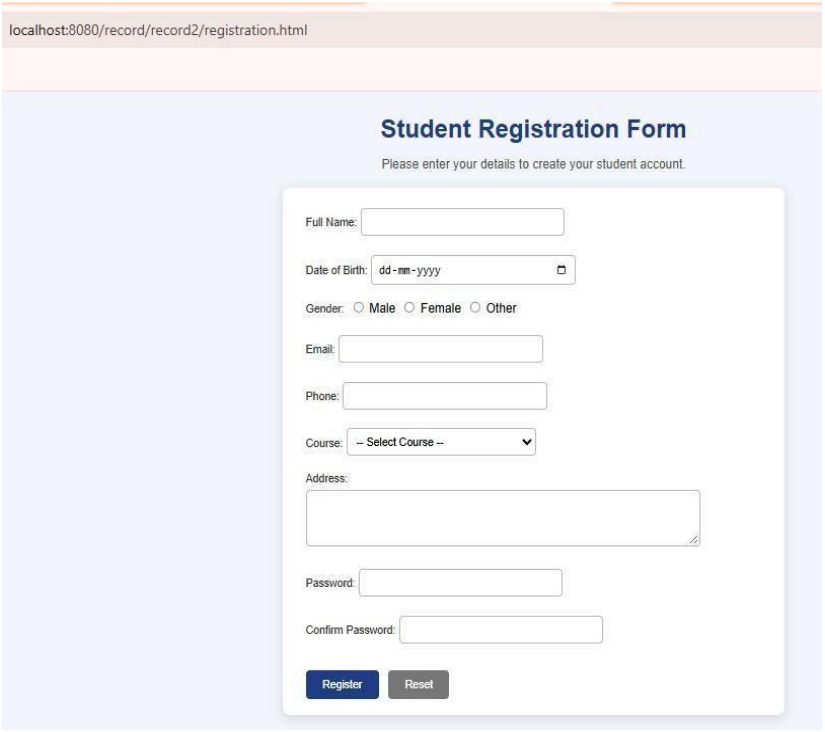
registration.css

```
 body {
    font-family: Arial, sans-serif;
    background-color: #f2f6fc;
    margin: 0;
    padding: 20px;
}
h1 {
    text-align: center;
    color: #1f3c88;
    margin-bottom: 5px;
}
body > p {
    text-align: center;
    color: #555;
    margin-bottom: 20px;
}
form {
    width: 600px;
    margin: auto;
    background-color: white;
    padding: 20px 30px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
}
label {
    font-size: 14px;
    color: #333;
}
input[type="text"],
input[type="email"],
input[type="tel"],
input[type="date"],
input[type="password"],
```

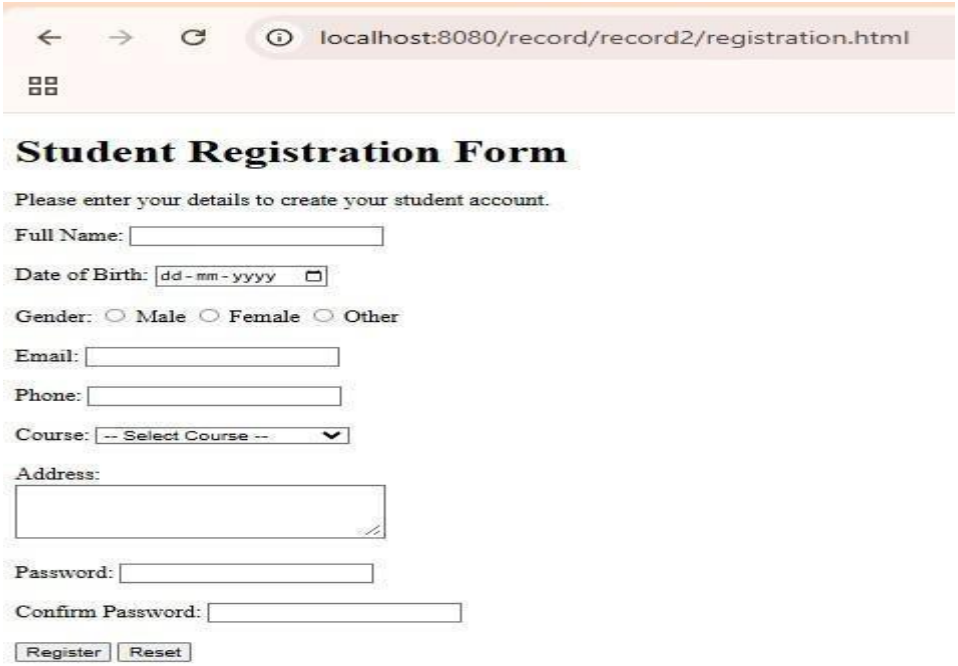
```
select {
    width: 250px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
}
textarea {
    width: 500px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
    resize: vertical;
}
input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #1f3c88;
    box-shadow: 0 0 5px rgba(31, 60, 136, 0.3);
}
input[type="radio"] {
    margin-left: 8px;
}
input[type="submit"],
input[type="reset"] {
    padding: 10px 22px;
    border: none;
    border-radius: 5px;
    font-size: 14px;
    cursor: pointer;
    margin-top: 15px;
}
input[type="submit"] {
    background-color: #1f3c88;
    color: white;
}
input[type="reset"] {
    background-color: #777;
    color: white;
    margin-left: 8px;
}
input[type="submit"]:hover,
input[type="reset"]:hover {
    opacity: 0.85;}
```

OUTPUT:

Positive Scenario – Student Registration Form with CSS Styling



Negative Scenario – Student Registration Form without CSS Styling



RESULT:

Thus, a Student Registration Form was successfully created using different HTML input types and form elements and styled using CSS.

5	**CLIENT VALIDATION - JAVASCRIPT: Login Form**	05Aug2026

AIM:
To familiarize students with JAVASCRIPT event handling and form validation

CONTEXT:
JavaScript is a scripting language used to add interactivity and dynamic behavior to web pages. In this program, JavaScript is used to validate the email and password fields before submitting the Login Form.

JavaScript Concepts Used:

JavaScript Concept	**Purpose**
document.getElementById()	Accesses an HTML element using its ID.
addEventListener()	Handles events such as form submission.
value	Gets the value entered in an input field.
trim()	Removes unnecessary spaces from the beginning and end of text.
if statement	Checks whether a condition is satisfied.
alert()	Displays a message to the user.
preventDefault()	Prevents form submission when validation fails.
Regular Expression	Checks whether the entered email follows the required format.

SOURCE CODE:

login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
   <link rel="stylesheet" href="login.css">
</head>
<body>
<center>
<form id="loginForm" method="post">
   <h1>Login</h1>
   <label>Email ID:</label>
   <input type="email" id="email">
   <span id="emailError"></span>
   <br><br>
   <label>Password:</label>
   <input type="password" id="password">
   <span id="passwordError"></span>
   <br><br>
   <input type="submit" value="Login">
   <input type="reset" value="Clear">
</form>
</center>
<script src="login.js"></script>
</body>
</html>
```

login.css

```
body {
   background-color: white;
   font-family: Arial;
}
form {
   width: 350px;
   padding: 25px;
   margin: 120px auto;
   background-color: white;
   border: 2px solid violet;
   border-radius: 10px;
}
h1 {
   color: purple;
}
input {
   padding: 7px;
```

```
    margin: 5px;
}
input[type="submit"],
input[type="reset"] {
    background-color: violet;
    color: white;
    border: none;
    border-radius: 5px;
}
span {
    color: red;
    font-size: 12px;
}
```

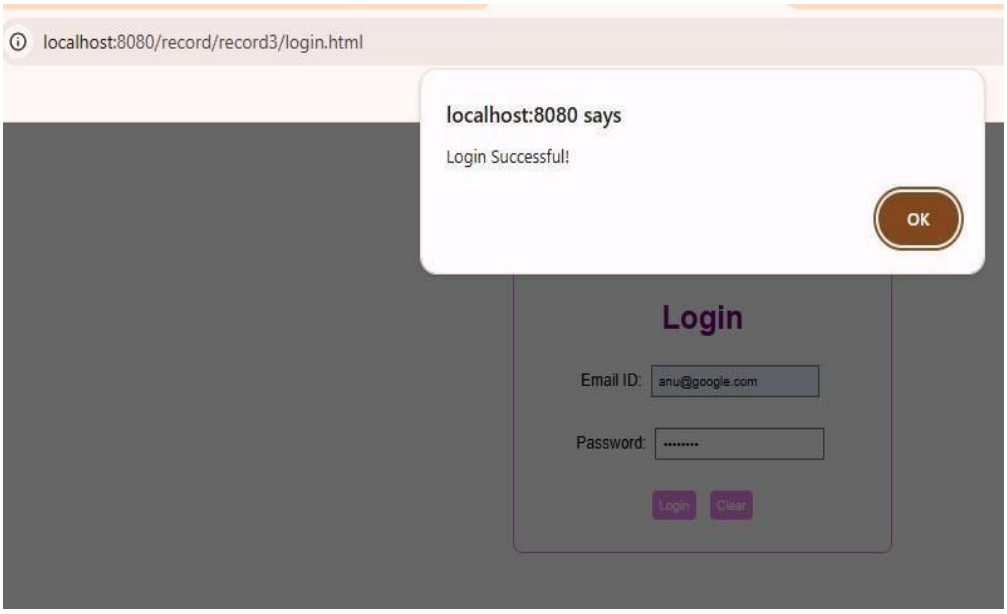
Login.js

```
let email = document.getElementById("email");
let password = document.getElementById("password");
function checkPassword() {
    let p = password.value;
    if (p.length < 8)
        passwordError.innerHTML = "Minimum 8 characters";
    else if (!/[A-Z]/.test(p))
        passwordError.innerHTML = "Add uppercase";
    else if (!/[a-z]/.test(p))
        passwordError.innerHTML = "Add lowercase";
    else if (!/[0-9]/.test(p))
        passwordError.innerHTML = "Add number";
    else if (!/[!@#$%^&*]/.test(p))
        passwordError.innerHTML = "Add special character";
    else
        passwordError.innerHTML = "";
}
email.addEventListener("blur", function() {
    emailError.innerHTML =
        email.value.includes("@") ? "" : "Email must include @";
});
password.addEventListener("blur", checkPassword);
loginForm.addEventListener("submit", function(event) {
    checkPassword();
    let p = password.value;
    if (email.value == "" || !email.value.includes("@") ||
        p.length < 8 || !/[A-Z]/.test(p) ||
        !/[a-z]/.test(p) || !/[0-9]/.test(p) ||
        !/[!@#$%^&*]/.test(p)) {
        event.preventDefault();
        alert("Enter a valid email and password");}
    else {
        event.preventDefault(); alert("Login Successful!");}});
```

OUTPUT:

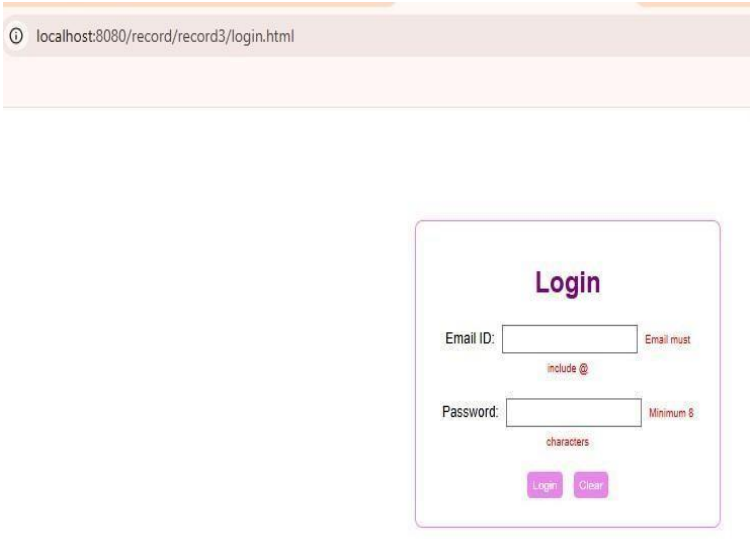
Positive Scenario – Login Form with Valid Details

The user enters a valid email address and password, and the login is successfully processed.



Negative Scenario – Login Form with Invalid Details

The user leaves the fields empty or enters invalid login details, and an appropriate validation message is displayed.



RESULT:

Thus, a Login Form was successfully validated using JAVASCRIPT with appropriate event handling and input validation.

6	**CLIENT VALIDATION - JAVASCRIPT: Registration Form**	10Aug2026

AIM:
To familiarize students with JAVASCRIPT form validation and event handling, and to validate the details entered in a Student Registration Form.

CONTEXT:
JavaScript is used in this program to validate the student details before the form is submitted. It checks the name, date of birth, gender, email, phone number, course, address, password, and confirm password fields. JavaScript Concepts Used:

JavaScript Concept	Purpose
getElementById()	Accesses form elements using their IDs.
querySelector()	Selects the required form element.
addEventListener()	Handles form submission and reset events.
value	Retrieves the entered value.
trim()	Removes unnecessary spaces from text fields.
if statement	Performs validation checks.
Regular Expression	Validates name, email, phone, and password formats.
focus()	Places the cursor on the field requiring correction.
alert()	Displays validation messages.
preventDefault()	Prevents submission when validation fails.
form.submit()	Submits the form after successful validation.

SOURCE CODE:

registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Student Registration</title>
   <link rel="stylesheet" href="registration.css">
</head>
<body>
   <h1>Student Registration Form</h1>
   <p>Please enter your details to create your student account.</p>
  <form id="registrationForm">
      <label for="name">Full Name:</label>        <input type="text"
id="name" name="name"        oninput="this.value
= this.value.replace(/[^A-Za-z ]/g, '')">
      <br><br>
      <label for="dob">Date of Birth:</label>
      <input type="date" id="dob" name="dob">
      <br><br>
      <label>Gender:</label>
      <input type="radio" name="gender" value="Male"> Male
      <input type="radio" name="gender" value="Female"> Female
      <input type="radio" name="gender" value="Other"> Other
      <br><br>
      <label for="email">Email:</label>
      <input type="email" id="email" name="email">
      <br><br>
      <label for="phone">Phone:</label>            <input
type="tel" id="phone" name="phone"
maxlength="10"       oninput="this.value =
this.value.replace(/[^0-9]/g, '')">
      <br><br>
      <label for="course">Course:</label>
      <select id="course" name="course">
        <option value="">-- Select Course --</option>
        <option>BCA</option>
        <option>B.Sc Computer Science</option>
        <option>B.Sc IT</option>
        <option>B.Com</option>
      </select>
      <br><br>
      <label for="address">Address:</label>
<br>
```

```
            <textarea id="address" name="address" rows="3" cols="30"></textarea>
            <br><br>
            <label for="password">Password:</label>
            <input type="password" id="password" name="password"> <br><br>
            <label for="confirmPassword">Confirm Password:</label>
            <input type="password" id="confirmPassword" name="confirmPassword">
            <br><br>
            <input type="submit" value="Register">
            <input type="reset" value="Reset">
        </form>
        <script src="registration.js"></script>
    </body>
    </html>
```

Registration.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f6fc;
    margin: 0;
    padding: 20px;
}
h1 {
    text-align: center;
    color: #1f3c88;
    margin-bottom: 5px;
}
body > p {
    text-align: center;
    color: #555;
    margin-bottom: 20px;
}
form {
    width: 600px;
    margin: auto;
    background-color: white;
    padding: 20px 30px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
}
label {
    font-size: 14px;
    color: #333;
}
input[type="text"],
input[type="email"],
input[type="tel"],
input[type="date"],
```

```
input[type="password"],
select {
    width: 250px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
}
textarea {
    width: 500px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
    resize: vertical;
}
input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #1f3c88;
    box-shadow: 0 0 5px rgba(31, 60, 136, 0.3);
}
input[type="radio"] {
    margin-left: 8px;
}
input[type="submit"],
input[type="reset"] {
    padding: 10px 22px;
    border: none;
    border-radius: 5px;
    font-size: 14px;
    cursor: pointer;
    margin-top: 15px;
}
input[type="submit"] {
    background-color: #1f3c88;
    color: white;
}
input[type="reset"] {
    background-color: #777;
    color: white;
    margin-left: 8px;
}
input[type="submit"]:hover,
input[type="reset"]:hover {
```

```
    opacity: 0.85;
}
```

registration.js

```
document.addEventListener("DOMContentLoaded", function () {
    const form = document.getElementById("registrationForm");
    form.addEventListener("submit", function (event) {
        event.preventDefault();
        // Get values
        // trim() is used except for Course, Password and Confirm Password
        const name = document.getElementById("name").value.trim();
        const dob = document.getElementById("dob").value.trim();
        const email = document.getElementById("email").value.trim();
        const phone = document.getElementById("phone").value.trim();
        const course = document.getElementById("course").value;
        const address = document.getElementById("address").value.trim();
        const password = document.getElementById("password").value;
        const confirmPassword =
            document.getElementById("confirmPassword").value;
        // Get selected gender
        const gender = document.querySelector(
            'input[name="gender"]:checked'
        );
        // ----------------------------------
        // 1. FULL NAME VALIDATION
        // ----------------------------------
        if (name === "") {
            alert("Please enter your full name.");
            document.getElementById("name").focus();
            return;
        }
        // Only letters and spaces are allowed
        const namePattern = /^[A-Za-z ]+$/;
        if (!namePattern.test(name)) {
            alert("Full Name should contain only letters and spaces.");
            document.getElementById("name").focus();
            return;
        }
        // ----------------------------------
        // 2. DATE OF BIRTH VALIDATION
        // ----------------------------------
        if (dob === "") {
            alert("Please select your Date of Birth.");
            document.getElementById("dob").focus();
            return;
        }
        const selectedDate = new Date(dob);
        const today = new Date();
```

```
today.setHours(0, 0, 0, 0);
// Future date check
if (selectedDate > today) {
    alert("Date of Birth cannot be a future date.");
    document.getElementById("dob").focus();
    return;
}
// Latest accepted birth year is 2011
const birthYear = selectedDate.getFullYear();
if (birthYear > 2011) {
    alert(
        "Students born after 2011 are not eligible for registration."
    );
    document.getElementById("dob").focus();
    return;
}
// ------------------------------------
// 3. GENDER VALIDATION
// ------------------------------------
if (!gender) {
    alert("Please select your gender.");
    return;
}
// ------------------------------------
// 4. EMAIL VALIDATION
// ------------------------------------
if (email === "") {
    alert("Please enter your email address.");
    document.getElementById("email").focus();
    return;
}
const emailPattern =
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
if (!emailPattern.test(email)) {
    alert("Please enter a valid email address.");
    document.getElementById("email").focus();
    return;
}
// ------------------------------------
// 5. PHONE VALIDATION
// ------------------------------------
if (phone === "") {
    alert("Please enter your phone number.");
    document.getElementById("phone").focus();
    return;
}
const phonePattern = /^[0-9]{10}$/;
if (!phonePattern.test(phone)) {
```

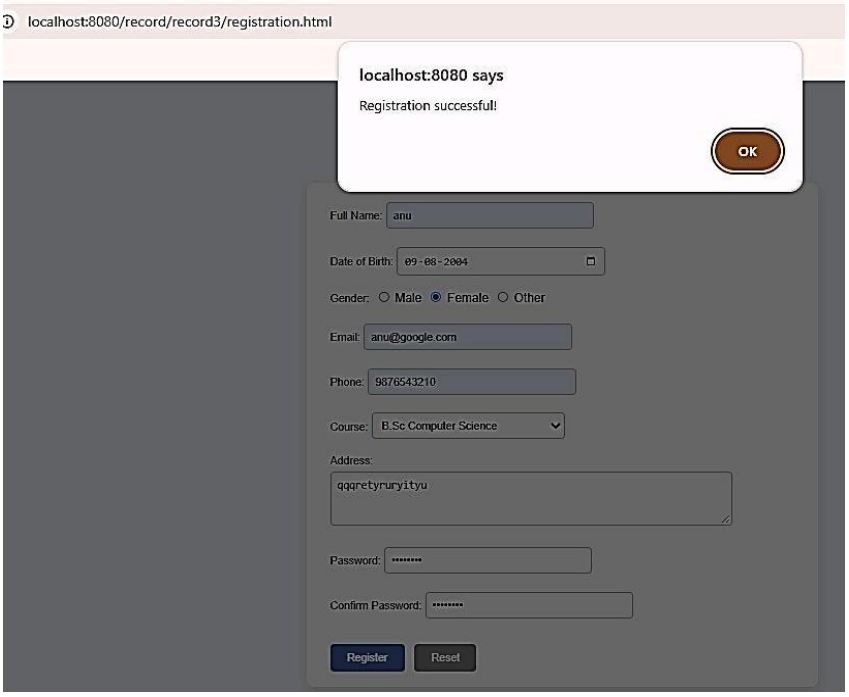
```
    alert("Phone number must contain exactly 10 digits.");
    document.getElementById("phone").focus();
    return;
}
// ----------------------------------
// 6. COURSE VALIDATION
// ----------------------------------
if (course === "") {
    alert("Please select a course.");
    document.getElementById("course").focus();
    return;
}
// ----------------------------------
// 7. ADDRESS VALIDATION
// ----------------------------------
if (address === "") {
    alert("Please enter your address.");
    document.getElementById("address").focus();
    return;
}
if (address.length < 10) {
    alert("Please enter a complete address.");
    document.getElementById("address").focus();
    return;
}
// ----------------------------------
// 8. PASSWORD VALIDATION
// ----------------------------------
if (password === "") {
    alert("Please enter a password.");
    document.getElementById("password").focus();
    return;
}
if (password.length < 8) {
    alert("Password must contain at least 8 characters.");
    document.getElementById("password").focus();
    return;
}
// Password must contain:
// uppercase + lowercase + number + special character
const passwordPattern =
    /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&]).+$/;
if (!passwordPattern.test(password)) {
    alert(
        "Password must contain at least one uppercase letter, " +
        "one lowercase letter, one number, and one special character."
    );
    document.getElementById("password").focus();
```

```
            return;
        }
        // ----------------------------------
        // 9. CONFIRM PASSWORD VALIDATION
        // ----------------------------------
        if (confirmPassword === "") {
            alert("Please confirm your password.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        if (password !== confirmPassword) {
            alert("Password and Confirm Password do not match.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        // ----------------------------------
        // 10. SUCCESS
        // ----------------------------------
        alert("Registration successful!");
        form.reset();
    });
    // ----------------------------------
    // RESET BUTTON
    // ----------------------------------
    form.addEventListener("reset", function () {
        setTimeout(function () {
            document.getElementById("name").focus();
        }, 0);

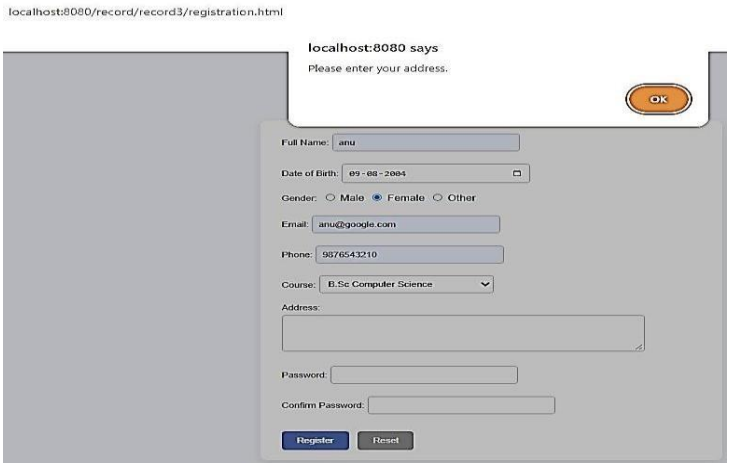
    });
});
```

OUTPUT:

Positive Scenario – Student Registration Form with Valid Details



Negative Scenario – Student Registration Form with Invalid Details



RESULT:

Thus, a Student Registration Form was successfully validated using JAVASCRIPT with appropriate input validation and event handling.

7	**SERVER-SIDE SCRIPTING: Login Using PHP**	02Sep2026

AIM:

To familiarize students with PHP and server-side form processing by accepting and validating user login credentials.

CONTEXT:

PHP (Hypertext Preprocessor) is a server-side scripting language used to create dynamic web pages. In this program, PHP is used to receive the registration form data, connect to the MySQL database, insert the student details, and display the registered details.

PHP CONCEPTS USED

PHP Concept	Description
<?php ?>	Defines a PHP code block.
$_POST	Retrieves data submitted through a POST form.
include	Includes another PHP file such as the database connection file.
mysqli	Provides a connection between PHP and MySQL.
prepare()	Prepares an SQL statement for execution.
bind_param()	Binds values to the prepared SQL statement.
execute()	Executes the prepared SQL statement.
if statement	Performs conditional checking.
htmlspecialchars()	Safely displays user-entered data in HTML.

SOURCE CODE:

login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
   <link rel="stylesheet" href="login.css">
</head>
<body>
<form id="loginForm" action="login.php" method="post" autocomplete="off">
<h1>Login</h1>
```

```
        <label>Email ID:</label>
        <input type="email" id="email" name="email" autocomplete="off">
        <span id="emailError"></span>
        <br><br>
        <label>Password:</label>
        <input type="password" id="password" name="password"
     placeholder="Minimum 8 characters"          autocomplete="newpassword">
        <span id="passwordError"></span>
        <br><br>
        <input type="submit" value="Login">
        <input type="reset" value="Clear">
     </form>
     <script src="login.js"></script>
     </body>
     </html>
```

Login.css

```
body {
   background-color: white;
   font-family: Arial;
}
form {
   width: 350px;
   padding: 25px;
   margin: 120px auto;
   background-color: white;
   border: 2px solid violet;
   border-radius: 10px;
}
h1 {
   text-align: center;
   color: purple;
}
input {
   padding: 7px;
   margin: 5px;
}
input[type="submit"],
input[type="reset"] {
   background-color: violet;
   color: white;
   border: none;
   border-radius: 5px;
}
span {
   color: red;
```

```
    font-size: 12px;
}
.result {
    width: 350px;
    padding: 15px;
    margin: 30px auto;
    border: 2px solid violet;
    border-radius: 10px;
    text-align: center;
}
.result h2 {
    color: purple;
}
```

Login.js

```
let email = document.getElementById("email");
let password = document.getElementById("password");
function checkPassword() {
    let p = password.value;
    if (p.length < 8)
        passwordError.innerHTML = "Minimum 8 characters";
    else if (!/[A-Z]/.test(p))
        passwordError.innerHTML = "Add uppercase";
    else if (!/[a-z]/.test(p))
        passwordError.innerHTML = "Add lowercase";
    else if (!/[0-9]/.test(p))
        passwordError.innerHTML = "Add number";
    else if (!/[!@#$%^&*]/.test(p))
        passwordError.innerHTML = "Add special character";
    else
        passwordError.innerHTML = "";
}
email.addEventListener("blur", function() {
    if (email.value == "")
        emailError.innerHTML = "Email is required";
    else if (!email.value.includes("@"))
        emailError.innerHTML = "Email must include @";
    else
        emailError.innerHTML = "";
});
password.addEventListener("blur", checkPassword);
loginForm.addEventListener("submit", function(event) {
    checkPassword();
    let p = password.value;
    if (email.value == "" ||
        !email.value.includes("@") ||
        p.length < 8 ||
```

```
        !/[A-Z]/.test(p) ||
        !/[a-z]/.test(p) ||
        !/[0-9]/.test(p) ||
        !/[!@#$%^&*]/.test(p)) {
        event.preventDefault();
        alert("Enter valid email and password");
    }
});
```

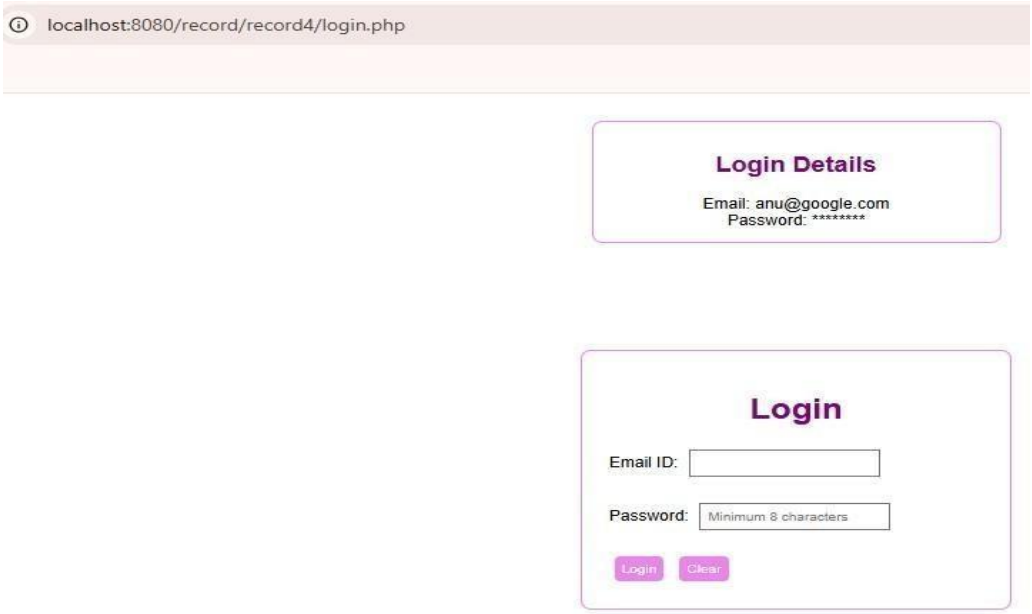
login.php

```
<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $email = $_POST["email"];
    $password = $_POST["password"];

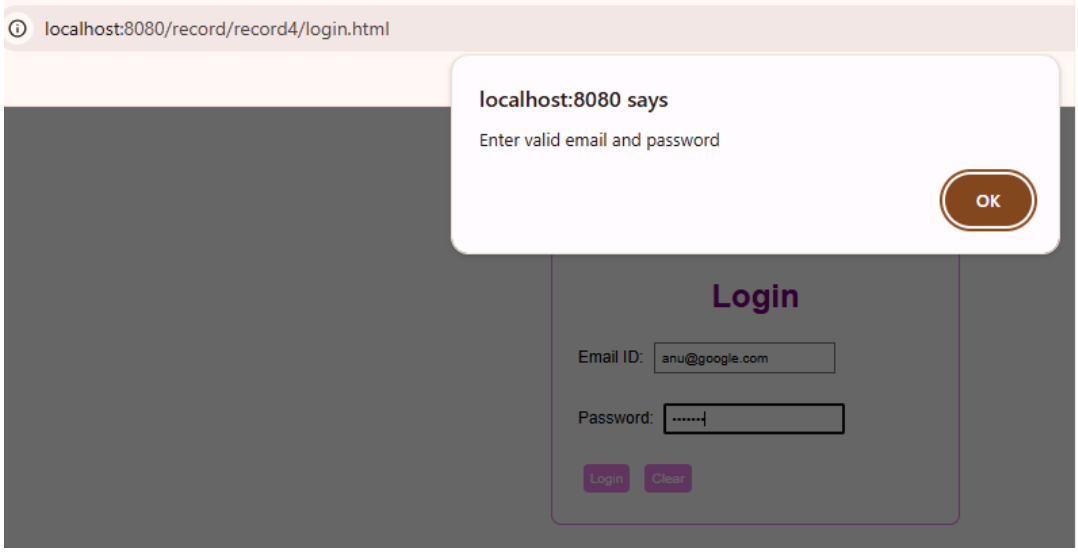
    echo "<div class='result'>";
    echo "<h2>Login Details</h2>";
    echo "Email: " . htmlspecialchars($email) . "<br>";
    echo "Password: " . htmlspecialchars($password);
    echo "</div>";
}
include "login.html";
?>
```

OUTPUT:

Positive Scenario – Student Registration with Valid Details
The student enters valid registration details. PHP processes the submitted data, stores it in the MySQL database, and displays the registered student details.



Negative Scenario – Registration with Invalid or Empty Details
The registration form is not processed when required details are missing or invalid. The user is prompted to correct the entered information before submission.



RESULT:
Thus, a Student Registration Form was successfully processed using PHP, and the student details were stored in and retrieved from the MySQL database.

8	**SERVER-SIDE SCRIPTING: Registration Form Using PHP**	07Sep2026

AIM:
To familiarize students with PHP server-side scripting and process Student Registration Form data using PHP.

CONTEXT:
PHP is a server-side scripting language used to process data submitted through web forms. In this program, PHP receives the student registration details using the POST method, performs basic processing, and displays the submitted details.

PHP CONCEPTS USED

PHP Concept	Description
<?php ?>	Defines a PHP code block.
$_POST	Retrieves data submitted from a form using the POST method.
include	Includes another PHP file in the current program.
if	Checks a specified condition.
isset()	Checks whether a variable or form value is available.
htmlspecialchars()	Safely displays user-entered data in HTML.
echo	Displays output on the webpage.

SOURCE CODE:
registration.php

```
<?php
$name = "";
$dob = "";
$gender = "";
$email = "";
$phone = "";
$course = "";
$address = "";
$submitted = false;
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = trim($_POST["name"] ?? "");
    $dob = trim($_POST["dob"] ?? "");
    $gender = $_POST["gender"] ?? "";
    $email = trim($_POST["email"] ?? "");
    $phone = trim($_POST["phone"] ?? "");
    $course = $_POST["course"] ?? "";
    $address = trim($_POST["address"] ?? "");
    $submitted = true;
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
        content="width=device-width, initial-scale=1.0">
    <title>Student Registration</title>
    <link rel="stylesheet" href="registration.css">
    <style>
        /* Main layout */
        .main-container {
            display: flex;
            justify-content: center;
            align-items: flex-start;
            gap: 25px;
            margin: 15px auto 30px;
            max-width: 1050px;
        }
        /* Registration form box */
        .form-box {
            width: 470px;
            background-color: white;
            padding: 20px 22px;
            border-radius: 10px;
            box-shadow: 0 4px 15px rgba(0, 0, 0, 0.12);
            box-sizing: border-box;
        }
        /* Result box */
        .result-box {
            width: 420px;
            background-color: white;
            padding: 20px 22px;
            border-radius: 10px;
            box-shadow: 0 4px 15px rgba(0, 0, 0, 0.12);
            box-sizing: border-box;
        }
        /* Result heading */
```

```
        .result-box h2 {
            text-align: center;
            color: #1f3c88;
            margin: 0 0 15px 0;
            font-size: 22px;
        }
        /* Result table */
        .result-table {
            width: 100%;
            border-collapse: collapse;
            font-size: 14px;
        }
        .result-table td {
            border: 1px solid #d9d9d9;
            padding: 8px 10px;
            vertical-align: top;
        }
        .result-table td:first-child {
            width: 38%;
            font-weight: bold;
            background-color: #f2f6fc;
            color: #333;
        }
        /* Keep address from making the box too large */
        .address-cell {
            word-break: break-word;
        }
        /* Responsive layout */
        @media (max-width: 900px) {
            .main-container {
                flex-direction: column;
                align-items: center;
            }
            .form-box,
            .result-box {
                width: 90%;
                max-width: 500px;
            }
        }
    </style>
</head>
<body>
    <h1>Student Registration Form</h1>
    <p>Please enter your details to create your student account.</p>
    <div class="main-container">
        <!-- REGISTRATION FORM -->
        <div class="form-box">
            <form id="registrationForm"
```

```
 method="POST">
<label for="name">Full Name:</label>
<input type="text"
      id="name"
      name="name"
      value="<?php echo htmlspecialchars($name); ?>"
      oninput="this.value = this.value.replace(/[^A-Za-z ]/g, '')">
<br><br>
<label for="dob">Date of Birth:</label>
<input type="date"
      id="dob"
      name="dob"
      value="<?php echo htmlspecialchars($dob); ?>">
<br><br>
<label>Gender:</label>
<input type="radio"
      name="gender"
      value="Male"
      <?php if ($gender == "Male") echo "checked"; ?>>
Male
<input type="radio"
      name="gender"
      value="Female"
      <?php if ($gender == "Female") echo "checked"; ?>>
Female
<input type="radio"
      name="gender"
      value="Other"
      <?php if ($gender == "Other") echo "checked"; ?>>
Other
<br><br>
<label for="email">Email:</label>
<input type="email"
      id="email"
      name="email"
      value="<?php echo htmlspecialchars($email); ?>">
<br><br>
<label for="phone">Phone:</label>
<input type="tel"
      id="phone"
      name="phone"
      maxlength="10"
      value="<?php echo htmlspecialchars($phone); ?>"
      oninput="this.value = this.value.replace(/[^0-9]/g, '')">
<br><br>
<label for="course">Course:</label>
<select id="course"
       name="course">
```

```
            <option value="">-- Select Course --</option>
            <option value="BCA"
               <?php if ($course == "BCA") echo "selected"; ?>>
               BCA
            </option>
            <option value="B.Sc Computer Science"
               <?php if ($course == "B.Sc Computer Science") echo "selected"; ?>>
               B.Sc Computer Science
            </option>
            <option value="B.Sc IT"
               <?php if ($course == "B.Sc IT") echo "selected"; ?>>
               B.Sc IT
            </option>
            <option value="B.Com"
               <?php if ($course == "B.Com") echo "selected"; ?>>
               B.Com
            </option>
         </select>
         <br><br>
         <label for="address">Address:</label>
         <br>
         <textarea id="address"
                  name="address"
                  rows="3"
                  cols="30"><?php echo htmlspecialchars($address); ?></textarea>
         <br><br>
         <label for="password">Password:</label>
         <input type="password"
               id="password"
               name="password">
         <br><br>
         <label for="confirmPassword">Confirm Password:</label>
         <input type="password"
               id="confirmPassword"
               name="confirmPassword">
         <br><br>
         <input type="submit"
               value="Register">
         <input type="reset"
               value="Reset">
      </form>
   </div>
   <!-- REGISTRATION DETAILS -->
   <?php if ($submitted): ?>
   <div class="result-box">
      <h2>Registration Details</h2>
      <table class="result-table">
         <tr>
```

```
            <td>Full Name</td>
            <td>
                <?php echo htmlspecialchars($name); ?>
            </td>
        </tr>
        <tr>
            <td>Date of Birth</td>
            <td>
                <?php
                if (!empty($dob)) {
                    echo date("d-m-Y", strtotime($dob));
                }
                ?>
            </td>
        </tr>
        <tr>
            <td>Gender</td>
            <td>
                <?php echo htmlspecialchars($gender); ?>
            </td>
        </tr>
        <tr>
            <td>Email</td>
            <td>
                <?php echo htmlspecialchars($email); ?>
            </td>
        </tr>
        <tr>
            <td>Phone</td>
            <td>
                <?php echo htmlspecialchars($phone); ?>
            </td>
        </tr>
        <tr>
            <td>Course</td>
            <td>
                <?php echo htmlspecialchars($course); ?>
            </td>
        </tr>
        <tr>
            <td>Address</td>
            <td class="address-cell">
                <?php echo htmlspecialchars($address); ?>
            </td>
        </tr>
    </table>
</div>
<?php endif; ?>
```

```
    </div>
    <script src="registration.js"></script>
</body>
</html>
registration.css
body {
    font-family: Arial, sans-serif;
    background-color: #f2f6fc;
    margin: 0;
    padding: 20px;
}
h1 {
    text-align: center;
    color: #1f3c88;
    margin-bottom: 5px;
}
body > p {
    text-align: center;
    color: #555;
    margin-bottom: 20px;
}
form {
    width: 600px;
    margin: auto;
    background-color: white;
    padding: 20px 30px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
}
label {
    font-size: 14px;
    color: #333;
}
input[type="text"],
input[type="email"],
input[type="tel"],
input[type="date"],
input[type="password"],
select {
    width: 250px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
}
textarea {
    width: 500px;
```

```
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
    resize: vertical;
}
input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #1f3c88;
    box-shadow: 0 0 5px rgba(31, 60, 136, 0.3);
}
input[type="radio"] {
    margin-left: 8px;
}
input[type="submit"],
input[type="reset"] {
    padding: 10px 22px;
    border: none;
    border-radius: 5px;
    font-size: 14px;
    cursor: pointer;
    margin-top: 15px;
}
input[type="submit"] {
    background-color: #1f3c88;
    color: white;
}
input[type="reset"] {
    background-color: #777;
    color: white;
    margin-left: 8px;
}
input[type="submit"]:hover,
input[type="reset"]:hover {
    opacity: 0.85;
}
```

registration.js

```
document.addEventListener("DOMContentLoaded", function () {
    const form = document.getElementById("registrationForm");
    form.addEventListener("submit", function (event) {
        event.preventDefault();
        // Get values
        const name = document.getElementById("name").value.trim();
        const dob = document.getElementById("dob").value.trim();
```

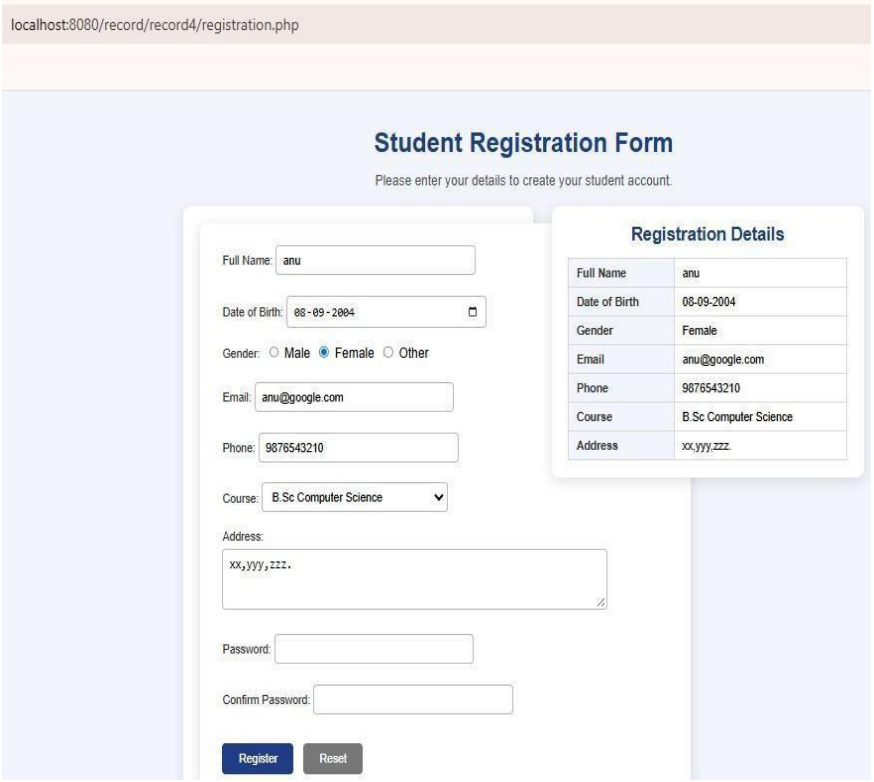
```
const email = document.getElementById("email").value.trim();
const phone = document.getElementById("phone").value.trim();
// Course, Password and Confirm Password are not trimmed
const course = document.getElementById("course").value;
const address = document.getElementById("address").value.trim();
const password = document.getElementById("password").value;
const confirmPassword =
   document.getElementById("confirmPassword").value;
// Gender
const gender = document.querySelector(
   'input[name="gender"]:checked'
);
// 1. FULL NAME
if (name === "") {
   alert("Please enter your full name.");
   document.getElementById("name").focus();
   return;
}
const namePattern = /^[A-Za-z ]+$/;
if (!namePattern.test(name)) {
   alert("Full Name should contain only letters and spaces.");
   document.getElementById("name").focus();
   return;
}
// 2. DATE OF BIRTH
if (dob === "") {
   alert("Please select your Date of Birth.");
   document.getElementById("dob").focus();
   return;
}
const selectedDate = new Date(dob);
const today = new Date();
today.setHours(0, 0, 0, 0);
if (selectedDate > today) {
   alert("Date of Birth cannot be a future date.");
   document.getElementById("dob").focus();
   return;
}
const birthYear = selectedDate.getFullYear();
if (birthYear > 2011) {
   alert(
      "Students born after 2011 are not eligible for registration."
   );
   document.getElementById("dob").focus();
   return;
}
// 3. GENDER
if (!gender) {
```

```
    alert("Please select your gender.");
    return;
}
// 4. EMAIL
if (email === "") {
    alert("Please enter your email address.");
    document.getElementById("email").focus();
    return;
}
const emailPattern =
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
if (!emailPattern.test(email)) {
    alert("Please enter a valid email address.");
    document.getElementById("email").focus();
    return;
}
// 5. PHONE
if (phone === "") {
    alert("Please enter your phone number.");
    document.getElementById("phone").focus();
    return;
}
const phonePattern = /^[0-9]{10}$/;
if (!phonePattern.test(phone)) {
    alert("Phone number must contain exactly 10 digits.");
    document.getElementById("phone").focus();
    return;
}
// 6. COURSE
if (course === "") {
    alert("Please select a course.");
    document.getElementById("course").focus();
    return;
}
// 7. ADDRESS
if (address === "") {
    alert("Please enter your address.");
    document.getElementById("address").focus();
    return;
}
if (address.length < 10) {
    alert("Please enter a complete address.");
    document.getElementById("address").focus();
    return;
}
// 8. PASSWORD
if (password === "") {
    alert("Please enter a password.");
```

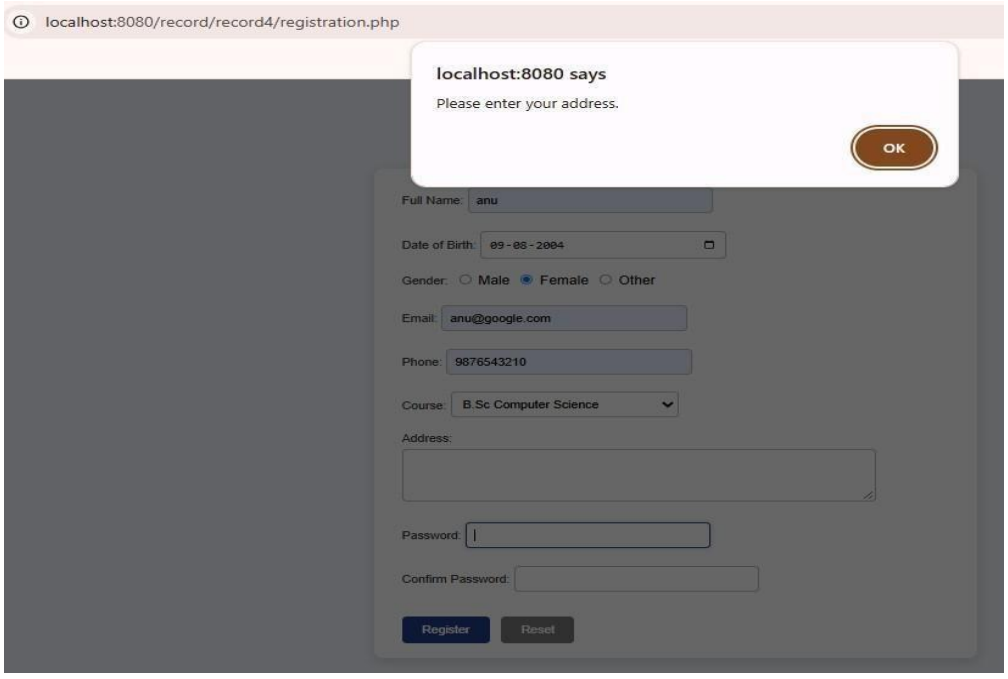
```
            document.getElementById("password").focus();
            return;
        }
        if (password.length < 8) {
            alert("Password must contain at least 8 characters.");
            document.getElementById("password").focus();
            return;
        }
        const passwordPattern =
            /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&]).+$/;
        if (!passwordPattern.test(password)) {
            alert(
                "Password must contain at least one uppercase letter, " +
                "one lowercase letter, one number, and one special character."
            );
            document.getElementById("password").focus();
            return;
        }
        // 9. CONFIRM PASSWORD
        if (confirmPassword === "") {
            alert("Please confirm your password.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        if (password !== confirmPassword) {
            alert("Password and Confirm Password do not match.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        // 10. ALL VALID
        alert("Registration successful!");
        // Send the form to PHP
        form.submit();
    });
    // RESET BUTTON
    form.addEventListener("reset", function () {
        setTimeout(function () {
            document.getElementById("name").focus();
        }, 0);
    });
});
```

OUTPUT:

Positive Scenario – Valid Student Registration



Negative Scenario – Empty or Invalid Details



RESULT:

Thus, a Student Registration Form was successfully processed using PHP, and the submitted student details were displayed using server-side scripting.

9	**PHP - DATABASE CONNECTIVITY: Registration Form**	16Sep2026

AIM:
To connect a PHP Student Registration Form with a MySQL database and store the submitted student details in a database table.

CONTEXT:
In this program, PHP is connected to a MySQL database to store student registration details. The submitted form data is received using PHP and inserted into the database using an SQL statement.

CONCEPTS USED:

Concept	Description
mysqli	Establishes a connection between PHP and MySQL.
new mysqli()	Creates a database connection.
connect_error	Checks whether the database connection failed.
set_charset()	Sets the character encoding for the database connection.
INSERT INTO	Inserts new records into a database table.
prepare()	Prepares an SQL statement for execution.
bind_param()	Binds form values to the SQL statement.
execute()	Executes the prepared SQL statement.
$_POST	Retrieves data submitted from the registration form.

SOURCE CODE:

Registration.php

```
<?php
include "db.php";
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    // Get values from form
    $name = trim($_POST["name"]);
    $dob = trim($_POST["dob"]);
    $gender = $_POST["gender"] ?? "";
    $email = trim($_POST["email"]);
    $phone = trim($_POST["phone"]);
    $course = $_POST["course"] ?? "";
    $address = trim($_POST["address"]);
    $password = $_POST["password"];
    $confirmPassword = $_POST["confirmPassword"];
    // Check password and confirm password
    if ($password !== $confirmPassword) {
        die("Passwords do not match.");
    }
    // Check whether email already exists
    $checkSql = "SELECT id FROM registration WHERE email = ?";

    $checkStmt = $conn->prepare($checkSql);
    $checkStmt->bind_param("s", $email);
    $checkStmt->execute();
    $checkResult = $checkStmt->get_result();
    if ($checkResult->num_rows > 0) {
    echo "<script>
            alert('Email already registered. Please use another email.');
            window.history.back();
          </script>";
    exit();
}
    $checkStmt->close();
    // Hash password before storing
    $hashedPassword = password_hash($password, PASSWORD_DEFAULT);
    // Insert registration details
    $sql = "INSERT INTO registration
            (name, dob, gender, email, phone, course, address, password)
            VALUES (?, ?, ?, ?, ?, ?, ?, ?)";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param(
        "ssssssss",
        $name,
        $dob,
        $gender,
        $email,
```

```
    $phone,
    $course,
    $address,
    $hashedPassword
);
if ($stmt->execute()) {
    echo "<!DOCTYPE html>";
    echo "<html>";
    echo "<head>";
    echo "<title>Registration Successful</title>";
    echo "<style>";
    echo "body {
            font-family: Arial;
            background-color: #f2f6fc;
            padding: 40px;
          }";
    echo ".box {
            width: 500px;
            margin: auto;
            background: white;
            padding: 30px;
            border-radius: 10px;
            text-align: center;
            box-shadow: 0 4px 15px rgba(0,0,0,0.15);
          }";
    echo "h1 {
            color: green;
          }";
    echo ".login {
            display: inline-block;
            margin-top: 20px;
            padding: 10px 20px;
            background-color: #1f3c88;
            color: white;
            text-decoration: none;
            border-radius: 5px;
          }";
    echo "</style>";
    echo "</head>";
    echo "<body>";
    echo "<div class='box'>";
    echo "<h1>Registration Successful!</h1>";
    echo "<p>Your details have been stored in the database.</p>";
    echo "<p><b>Name:</b> " . htmlspecialchars($name) . "</p>";
    echo "<p><b>Email:</b> " . htmlspecialchars($email) . "</p>";
    echo "<p><b>Course:</b> " . htmlspecialchars($course) . "</p>";
    echo "<a class='login' href='login.html'>Go to Login</a>";
    echo "</div>";
```

```
        echo "</body>";
        echo "</html>";
    } else {
        echo "Registration failed: " . $stmt->error;
    }
    $stmt->close();
    $conn->close();
    exit();
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
          content="width=device-width, initial-scale=1.0">
    <title>Student Registration</title>
    <link rel="stylesheet" href="registration.css">
</head>
<body>
    <h1>Student Registration Form</h1>
    <p>Please enter your details to create your student account.</p>
    <form id="registrationForm"
          action="registration.php"
          method="POST">
        <label for="name">Full Name:</label>
        <input type="text"
               id="name"
               name="name">
        <br><br>
        <label for="dob">Date of Birth:</label>
        <input type="date"
               id="dob"
               name="dob">
        <br><br>
        <label>Gender:</label>
        <input type="radio"
               name="gender"
               value="Male">
        Male
        <input type="radio"
               name="gender"
               value="Female">
        Female
        <input type="radio"
               name="gender"
               value="Other">
        Other
```

```
    <br><br>
    <label for="email">Email:</label>
    <input type="email"
           id="email"
           name="email">
    <br><br>
    <label for="phone">Phone:</label>
    <input type="tel"
           id="phone"
           name="phone"
           maxlength="10">
    <br><br>
    <label for="course">Course:</label>
    <select id="course"
            name="course">
       <option value="">-- Select Course --</option>
       <option>BCA</option>
       <option>B.Sc Computer Science</option>
       <option>B.Sc IT</option>
       <option>B.Com</option>
    </select>
    <br><br>
    <label for="address">Address:</label>
    <br>
    <textarea id="address"
              name="address"
              rows="3"
              cols="30"></textarea>
    <br><br>

    <label for="password">Password:</label>
    <input type="password"
           id="password"
           name="password">
    <br><br>
    <label for="confirmPassword">
       Confirm Password:
    </label>
    <input type="password"
           id="confirmPassword"
           name="confirmPassword">
    <br><br>
    <input type="submit"
           value="Register">
    <input type="reset"
           value="Reset">
</form>
<script src="registration.js"></script>
```

```
</body>
</html>
```

Registration.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f6fc;
    margin: 0;
    padding: 20px;
}
h1 {
    text-align: center;
    color: #1f3c88;
    margin-bottom: 5px;
}
body > p {
    text-align: center;
    color: #555;
    margin-bottom: 20px;
}
form {
    width: 600px;
    margin: auto;
    background-color: white;
    padding: 20px 30px;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
}
label {
    font-size: 14px;
    color: #333;
}
input[type="text"],
input[type="email"],
input[type="tel"],
input[type="date"],
input[type="password"],
select {
    width: 250px;
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
}
textarea {
    width: 500px;
```

```
    padding: 8px;
    border: 1px solid #bbb;
    border-radius: 5px;
    font-size: 14px;
    margin-top: 5px;
    resize: vertical;
}
input:focus,
select:focus,
textarea:focus {
    outline: none;
    border-color: #1f3c88;
    box-shadow: 0 0 5px rgba(31, 60, 136, 0.3);
}
input[type="radio"] {
    margin-left: 8px;
}
input[type="submit"],
input[type="reset"] {
    padding: 10px 22px;
    border: none;
    border-radius: 5px;
    font-size: 14px;
    cursor: pointer;
    margin-top: 15px;
}
input[type="submit"] {
    background-color: #1f3c88;
    color: white;
}
input[type="reset"] {
    background-color: #777;
    color: white;
    margin-left: 8px;
}
input[type="submit"]:hover,
input[type="reset"]:hover {
    opacity: 0.85;
}
```

Registration.js

```
document.addEventListener("DOMContentLoaded", function () {
    const form = document.getElementById("registrationForm");
    form.addEventListener("submit", function (event) {
        event.preventDefault();
        // Get values
        const name = document.getElementById("name").value.trim();
```

```
const dob = document.getElementById("dob").value.trim();
const email = document.getElementById("email").value.trim();
const phone = document.getElementById("phone").value.trim();
// Course, Password and Confirm Password are not trimmed
const course = document.getElementById("course").value;
const address = document.getElementById("address").value.trim();
const password = document.getElementById("password").value;
const confirmPassword =
   document.getElementById("confirmPassword").value;
// Gender
const gender = document.querySelector(
   'input[name="gender"]:checked'
);
// 1. FULL NAME
if (name === "") {
   alert("Please enter your full name.");
   document.getElementById("name").focus();
   return;
}
const namePattern = /^[A-Za-z ]+$/;
if (!namePattern.test(name)) {
   alert("Full Name should contain only letters and spaces.");
   document.getElementById("name").focus();
   return;
}
// 2. DATE OF BIRTH
if (dob === "") {
   alert("Please select your Date of Birth.");
   document.getElementById("dob").focus();
   return;
}
const selectedDate = new Date(dob);
const today = new Date();
today.setHours(0, 0, 0, 0);
if (selectedDate > today) {
   alert("Date of Birth cannot be a future date.");
   document.getElementById("dob").focus();
   return;
}
const birthYear = selectedDate.getFullYear();
if (birthYear > 2011) {
   alert(
      "Students born after 2011 are not eligible for registration."
   );
   document.getElementById("dob").focus();
   return;
}
// 3. GENDER
```

```
if (!gender) {
    alert("Please select your gender.");
    return;
}
// 4. EMAIL
if (email === "") {
    alert("Please enter your email address.");
    document.getElementById("email").focus();
    return;
}
const emailPattern =
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
if (!emailPattern.test(email)) {
    alert("Please enter a valid email address.");
    document.getElementById("email").focus();
    return;
}
// 5. PHONE
if (phone === "") {
    alert("Please enter your phone number.");
    document.getElementById("phone").focus();
    return;
}
const phonePattern = /^[0-9]{10}$/;
if (!phonePattern.test(phone)) {
    alert("Phone number must contain exactly 10 digits.");
    document.getElementById("phone").focus();
    return;
}
// 6. COURSE
if (course === "") {
    alert("Please select a course.");
    document.getElementById("course").focus();
    return;
}
// 7. ADDRESS
if (address === "") {
    alert("Please enter your address.");
    document.getElementById("address").focus();
    return;
}
if (address.length < 10) {
    alert("Please enter a complete address.");
    document.getElementById("address").focus();
    return;
}
// 8. PASSWORD
if (password === "") {
```

```
            alert("Please enter a password.");
            document.getElementById("password").focus();
            return;
        }
        if (password.length < 8) {
            alert("Password must contain at least 8 characters.");
            document.getElementById("password").focus();
            return;
        }
        const passwordPattern =
            /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&]).+$/;
        if (!passwordPattern.test(password)) {
            alert(
                "Password must contain at least one uppercase letter, " +
                "one lowercase letter, one number, and one special character."
            );
            document.getElementById("password").focus();
            return;
        }
        // 9. CONFIRM PASSWORD
        if (confirmPassword === "") {
            alert("Please confirm your password.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        if (password !== confirmPassword) {
            alert("Password and Confirm Password do not match.");
            document.getElementById("confirmPassword").focus();
            return;
        }
        // 10. ALL VALID
        alert("Registration successful!");
        // Send the form to PHP
        form.submit();
    });
    // RESET BUTTON
    form.addEventListener("reset", function () {
        setTimeout(function () {
            document.getElementById("name").focus();
        }, 0);
    });
});
```

Registration_table:

```
CREATE TABLE IF NOT EXISTS registration (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
```

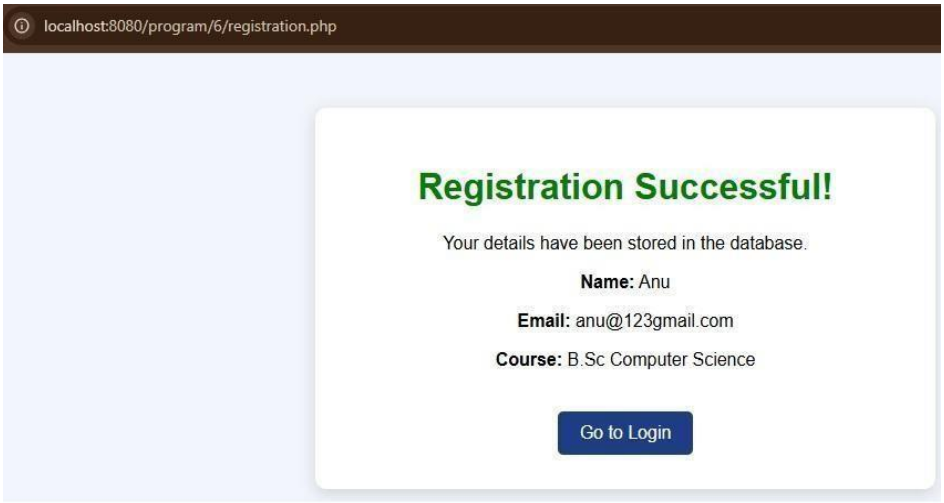
```
    dob DATE NOT NULL,
    gender VARCHAR(20) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(10) NOT NULL,
    course VARCHAR(100) NOT NULL,
    address TEXT NOT NULL,
    password VARCHAR(255) NOT NULL
);
```

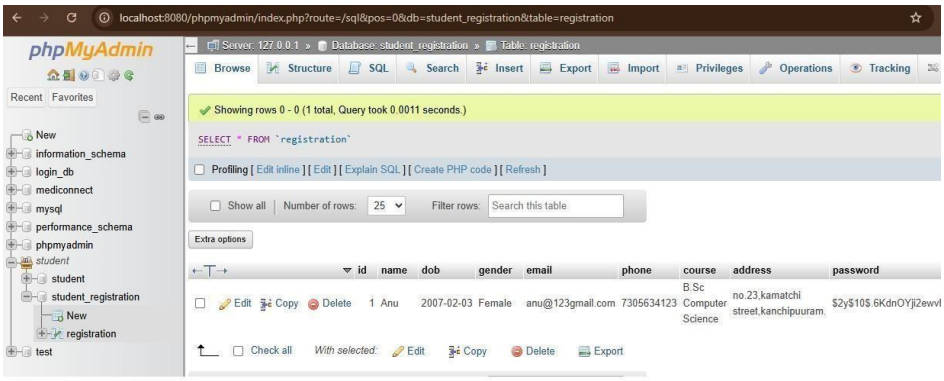
db.php

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$database = "student_registration";
$conn = new mysqli(
    $servername,
    $username,
    $password,
    $database
);
if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}
$conn->set_charset("utf8mb4");
?>
```

OUTPUT:

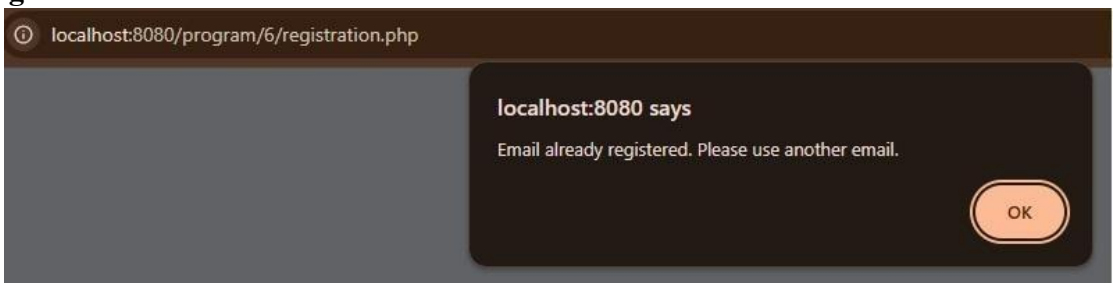
Positive Scenario – Successful Registration





When valid student details are submitted, PHP connects to the MySQL database and stores the registration details successfully.

Negative Scenario – Database Connection Failure



If the database connection fails, PHP displays a database connection error instead of storing the registration details.

RESULT :

Thus, the Student Registration Form was successfully connected to a MySQL database using PHP, and the submitted student details were stored in the database.

10	**PHP - DATABASE CONNECTIVITY: Login Form**	21Sep2026

To connect a PHP Login Form with a MySQL database and authenticate a user using the stored login credentials.

CONTEXT:
In this program, PHP is connected to a MySQL database to verify the login details entered by the user. The submitted email and password are compared with the corresponding records stored in the database. Access is allowed only when the login credentials are valid.

CONCEPTS USED:

Concept	Description
Mysqli	Establishes a connection between PHP and MySQL.
$_POST	Retrieves login details submitted through the form.
prepare()	Prepares an SQL statement for execution.
bind_param()	Binds user input values to the SQL statement.
execute()	Executes the prepared SQL statement.
get_result()	Retrieves the result returned by the SQL query.
num_rows	Checks the number of records returned by the query.
If	Checks whether the login credentials are valid.
Echo	Displays the login result to the user.

SOURCE CODE:

Login.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
   <link rel="stylesheet" href="login.css">
</head>
<body>
<form id="loginForm" action="login.php" method="post" autocomplete="off">
   <h1>Login</h1>
   <label>Email ID:</label>
   <input type="email" id="email" name="email" autocomplete="off">
   <span id="emailError"></span>
   <br><br>
   <label>Password:</label>
   <input type="password" id="password" name="password"
        placeholder="Minimum 8 characters">
   <span id="passwordError"></span>
   <br><br>
   <input type="submit" value="Login">
   <input type="reset" value="Clear">
</form>
<script src="login.js"></script>
</body>
</html>
```

Login.php

```
<?php
include "db.php";
if ($_SERVER["REQUEST_METHOD"] == "POST") {
   $email = trim($_POST["email"]);
   $password = $_POST["password"];
   // Search for email in registration table
   $sql = "SELECT * FROM registration WHERE email = ?";
   $stmt = $conn->prepare($sql);
   $stmt->bind_param("s", $email);
   $stmt->execute();
   $result = $stmt->get_result();
   // Check whether email exists
   if ($result->num_rows == 1) {
      $user = $result->fetch_assoc();
      // Check entered password with stored hashed password
      if (password_verify($password, $user["password"])) {
         echo "<!DOCTYPE html>";
         echo "<html>";
         echo "<head>";
         echo "<title>Login Successful</title>";
```

```
    echo "<style>";
    echo "body {
            font-family: Arial;
            background-color: #f2f6fc;
            padding: 50px;
          }";
    echo ".box {
            width: 450px;
            margin: auto;
            background: white;
            padding: 30px;
            text-align: center;
            border-radius: 10px;
            box-shadow: 0 4px 15px rgba(0,0,0,0.15);
          }";
    echo "h1 {
            color: green;
          }";
    echo "</style>";
    echo "</head>";
    echo "<body>";
    echo "<div class='box'>";
    echo "<h1>Login Successful!</h1>";
    echo "<p>Welcome, "
        . htmlspecialchars($user["name"])
        . ".</p>";
    echo "<p>Email: "
        . htmlspecialchars($user["email"])
        . "</p>";
    echo "</div>";
    echo "</body>";
    echo "</html>";
} else {
    echo "<!DOCTYPE html>";
    echo "<html>";
    echo "<head>";
    echo "<title>Invalid Login</title>";
    echo "</head>";
    echo "<body>";
    echo "<h1 style='color:red; text-align:center;'>
            Invalid Login
          </h1>";
    echo "<p style='text-align:center;'>
            Incorrect password.
          </p>";
    echo "<p style='text-align:center;'>
            <a href='login.html'>Try Again</a>
          </p>";
```

```
            echo "</body>";
            echo "</html>";
        }
    } else {
        echo "<!DOCTYPE html>";
        echo "<html>";
        echo "<head>";
        echo "<title>Invalid Login</title>";
        echo "</head>";
        echo "<body>";
        echo "<h1 style='color:red; text-align:center;'>
                Invalid Login
              </h1>";
        echo "<p style='text-align:center;'>
                Email is not registered.
              </p>";
        echo "<p style='text-align:center;'>
                <a href='login.html'>Try Again</a>
              </p>";
        echo "</body>";
        echo "</html>";
    }
    $stmt->close();
    $conn->close();
}
?>
```

db.php

```
<?php
$conn = mysqli_connect("localhost", "root", "", "login_db");
if (!$conn) {
    die("Database Connection Failed");
}
echo "Database Connected Successfully";
?>
```

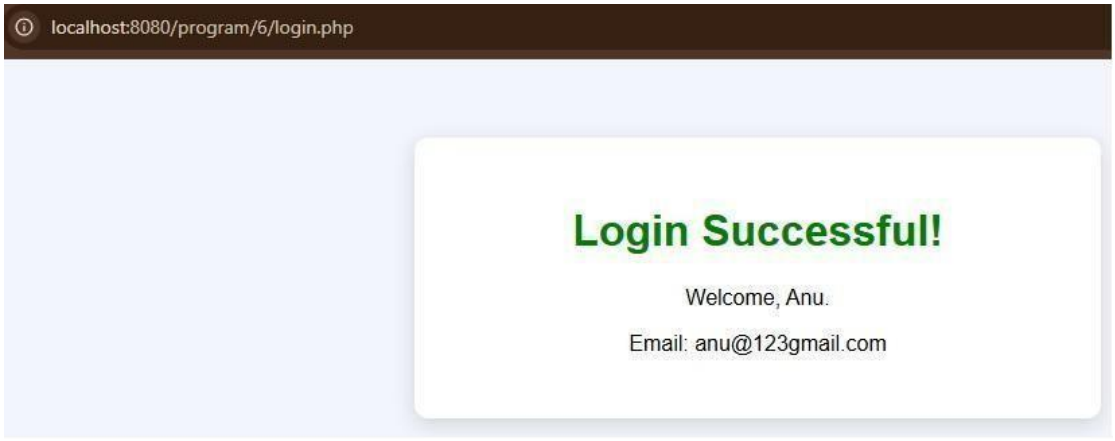
Display.php

```
<?php
include "db.php";
$sql = "SELECT * FROM users";
$result = mysqli_query($conn, $sql);
echo "<h2>Stored Login Details</h2>";
echo "<table border='1' cellpadding='10'>";
echo "<tr>";
echo "<th>ID</th>";
echo "<th>Email</th>";
echo "<th>Password</th>";
echo "</tr>";
```

```
while ($row = mysqli_fetch_assoc($result)) {
   echo "<tr>";
   echo "<td>" . $row["id"] . "</td>";
   echo "<td>" . htmlspecialchars($row["email"]) . "</td>";
   echo "<td>" . htmlspecialchars($row["password"]) . "</td>";
   echo "</tr>";
}
echo "</table>";
?>
```

OUTPUT:

Positive Scenario – Valid Login



When the user enters an email and password that match a record in the MySQL database, the login is successful and an appropriate success message is displayed.

Negative Scenario – Invalid Login



When the entered email or password does not match a stored record, the login is rejected and an appropriate error message is displayed.

RESULT:

Thus, the Login Form was successfully connected to a MySQL database using PHP, and user authentication was performed using the stored login credentials.